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Milestone 2

Risk Module Research & Design

Project Title: Design and Development of a Context-Aware Payment Decision Engine for Secure Voice-Based UPI Payments

VeriPay — "Verify Before You Pay."

1. Overview of Project

Voice-based UPI payments already exist — Razorpay, Sarvam — but none of them publicly demonstrate comprehensive real-time contextual risk analysis and explainable intent verification before payment execution. What none of them do is explain why they trust a voice command before executing it. This project is that missing decision layer: CPDE (Context-Aware Payment Decision Engine) pauses, reasons about intent and risk separately, and justifies its decision in plain language before money moves.

2. Milestone 2 Overview

Milestone 1 established the technical foundation — repository, backend environment, project structure and a working FastAPI server. Milestone 2 addresses the three research action items raised by the mentor in the Milestone 1 review meeting.

The main outcome of this milestone is a design change. The Payment Risk Score is no longer computed by a fixed set of hand-weighted rules. It is now produced by a trained machine-learning model that reads a much larger set of contextual features and explains each decision using SHAP feature attributions. The rule-based method has been retained, but only as a comparison baseline. This change is documented in a revised research proposal (v2), submitted alongside this report.

3. Action Items from Milestone 1 — Status

No.	Action Item Given by Mentor	Status	Addressed In
1	Research algorithms suitable for contextual Payment Risk Score analysis	Completed	Section 5
2	Identify additional risk signals that can improve CPDE's decision-making	Completed	Section 6
3	Study how existing payment systems perform risk assessment	Completed	Section 7

4. Objectives Achieved

- Researched and shortlisted five candidate approaches for Payment Risk Score computation, to be compared experimentally rather than chosen in advance.
- Expanded the risk signal set from 5 hand-weighted signals to approximately 25 engineered features grouped into four categories.
- Selected SHAP (SHapley Additive exPlanations) as the explainability method, so that every risk score can be traced to the specific features that produced it.
- Defined a three-layer dataset strategy, keeping general fraud data separate from voice-specific evaluation data.
- Defined a formal method for selecting the Proceed / Warn / Block thresholds from data, replacing the earlier hand-picked cut-offs.
- Stated the research contribution as a single testable hypothesis and designed a feature-category ablation study to test it.
- Reviewed how existing payment systems and published fraud-detection research perform risk assessment.
- Produced the revised research proposal document (VeriPay CPDE v2).

5. Algorithms Researched for Risk Score Analysis

Action item 1. Rather than committing to one algorithm now, five approaches were shortlisted and will be benchmarked against each other on the same data and the same metrics. The best performer becomes the risk model that feeds the decision policy.

Approach	Reason for Inclusion	Role in the Project
Rule-Based Scoring	The original Milestone-1 weighted-sum method; fully transparent by construction	Baseline to beat
Logistic Regression	Simple, fast, gives directly interpretable feature coefficients	Simple ML baseline
Random Forest	Handles non-linear feature interactions, trains quickly, not very sensitive to tuning	Strong practical candidate
XGBoost	Standard choice in published financial fraud detection; works well with SHAP	Primary candidate model
Graph Neural Network	Could model User - Account - Recipient - Merchant links	Conditional — only if the data shows real relational structure

Explainability. SHAP will be applied to the tree-based models (Random Forest, XGBoost) to break each risk score down into per-feature contributions. For example, a risk probability of 0.82 may be attributable to +0.25 from a new recipient, +0.21 from an unusual amount, +0.18 from urgency language in the spoken command and +0.10 from an unusual transaction time. CPDE converts these attributions into a plain-language sentence for the user instead of showing a raw number.

6. Expanded Contextual Risk Signals

Action item 2. The Milestone-1 design used 5 signals: recipient novelty, amount deviation, contact ambiguity, urgency language and high-amount pattern. These have been expanded into four feature categories covering roughly 25 features.

Category	Features
Transaction-level	Amount, transaction frequency, time of transaction, sudden high-value transaction, round-number amount, transaction velocity
User Behaviour	Deviation from normal spending, deviation from normal frequency, new recipient, new merchant, unusual transaction time, sudden change in behaviour
Recipient / Context	Recipient novelty, multiple matching contacts, previously unseen recipient, recipient transaction frequency, repeated failed or aborted attempts
Voice / Intent	Speech-to-text confidence, recipient extraction confidence, amount extraction confidence, ambiguous command, urgency or pressure language, contradictory information, repeated user corrections

The Voice / Intent category is the important one. It is what separates this project from a standard fraud-detection study. Existing fraud research works only on transaction tables. Adding features derived from how the command was spoken lets the risk model reason about the conversation itself, not just the money.

7. Study of Existing Payment Risk Assessment

Action item 3. Current UPI applications do carry out safety checks, but the checks are single-signal and static. A "new payee" warning appears whether the payment is Rs. 50 or Rs. 50,000, and the wording is the same every time. Published fraud-detection research, on the other hand, uses multi-feature learned models, but is evaluated on static tabular data and is not designed for real-time, voice-first, explainable interaction.

Property	Existing UPI Nudges	VeriPay (CPDE)
Signals used	One at a time (e.g. only "new payee")	Around 25 engineered features combined by a learned model
Scoring method	Fixed rule / static pop-up	Trained model, benchmarked against rule-based scoring
Explanation given	Generic ("new number, continue?")	SHAP-grounded, feature-specific, plain language
Compared to a baseline	No	Yes — rule-based vs. ML, on Precision / Recall / F1 / PR-AUC / FPR / latency

This confirms the research gap: no existing system combines real-time voice-command understanding, a learned multi-signal risk model, and a feature-grounded plain-language explanation in a single pipeline.

8. Dataset Strategy

Real UPI-native labelled transaction data is not available to a student project. Instead of relying on one small dataset, three separate layers are used, each testing a different part of the system. General fraud data is kept strictly separate from the voice-specific evaluation data.

Layer	Dataset	Purpose
A	PaySim (Lopez-Rojas et al., 2016) — about 6.3 million simulated mobile-money transactions, 0.13% fraud rate	Main dataset for training and evaluating the ML risk model; the closest available proxy to UPI-style transfers
A	IEEE-CIS Fraud Detection — 590,540 e-commerce transactions, 3.5% fraud rate	Cross-domain check that the model generalises
B	Own voice-command dataset, manually labelled Safe / Ambiguous / Risky	Evaluates the Speech - Intent - CPDE path specifically
C	10 hand-authored end-to-end scenarios covering Proceed / Warn / Block / Clarify	Full pipeline testing and viva demonstration

Stated openly: PaySim and IEEE-CIS are general and mobile-money fraud datasets, not UPI data. Results from Layer A therefore show that the risk model works in general, and are reported separately from the voice-specific results from Layers B and C. Both datasets are also heavily imbalanced (0.13% to 3.5% fraud), so class weighting or resampling will be required, and this affects how the metrics are read.

9. Threshold Selection Methodology

The Milestone-1 design used fixed cut-offs of 0.00–0.30 Proceed, 0.31–0.60 Warn and 0.61–1.00 Block. These were chosen by hand and cannot be defended as a result. They are now treated as a placeholder, and the final thresholds will be derived from data in four steps.

1. **Calibration.** Tree-based models rank transactions correctly, but their raw outputs are not true probabilities. The scores will be calibrated using Platt scaling or isotonic regression (sklearn.calibration.CalibratedClassifierCV) before being used as a Risk Score.
2. **Threshold search.** A grid search will be run over candidate (low, high) threshold pairs on the validation split, recording Precision, Recall, F1 and False Positive Rate at each pair.
3. **Cost-based objective.** The chosen pair is the one that minimises expected cost, where a missed risky payment is weighted more heavily than an unnecessary warning. The single assumption involved is the cost ratio between the two, which will be justified using published RBI / NPCI UPI fraud loss figures and reported alongside a sensitivity check showing how the thresholds move if that ratio changes.
4. **Validation, not test data.** The search is performed only on the validation split; the selected thresholds are then confirmed once on a held-out test split.

In short: the thresholds are not assumed. They are searched for on validation data using a cost objective, and the one judgement call involved — how much worse a missed scam is than a false alarm — is stated explicitly with a source rather than hidden inside a number.

10. Research Contribution and How It Will Be Tested

An important point was clarified during this milestone. Using real datasets such as PaySim with an established algorithm such as XGBoost makes the project rigorous, but it is not new work — that combination is already published. The contribution has therefore been narrowed to one sentence that can be proved or disproved:

> *Adding voice-derived features — speech-to-text confidence, recipient and amount extraction confidence, command ambiguity and urgency language — to standard transaction and behavioural features measurably improves risk detection compared with transaction-only models, while remaining explainable through SHAP.*

This is tested directly by a feature-category ablation study. The best model from Section 5 is retrained four times, adding one feature category each time, and the metrics are compared at every step:

Step	Features Used	What It Shows
1	Transaction-level only	Baseline detection performance
2	+ User Behaviour	Value added by spending-pattern history
3	+ Recipient / Context	Value added by who is being paid
4	+ Voice / Intent	The contribution claim — any improvement here is direct evidence that voice signals help

If step 4 shows no improvement, the hypothesis is disproved, and that is also a reportable result. The point is that the claim is falsifiable instead of assumed.

11. Evaluation Plan

Evaluation is organised in three tiers, so that each part of the system is measured against a fair comparison point.

Tier	Comparison	Metrics
Pipeline-level	Plain voice-payment flow vs. full VeriPay flow (Datasets B and C)	Accuracy, Precision, Recall, F1, confusion matrix, False Positive Rate, average decision time
Risk-model-level	Rule-based vs. Logistic Regression vs. Random Forest vs. XGBoost (Dataset A)	Precision, Recall, F1, PR-AUC, False Positive Rate, detection latency, explainability
Feature ablation	Same model retrained with one extra feature category at a time	F1, Recall and PR-AUC compared at each of the four steps

12. Key Finding During This Milestone

While designing the ablation study, a methodological problem was identified and corrected before any code was written.

The problem. Three of the Voice / Intent features — speech-to-text confidence and the two extraction confidence values — are outputs of a real speech recognition engine running on real recorded audio. If those numbers are generated synthetically, the ablation becomes circular: the generator would decide how strongly voice confidence relates to risk, and the experiment would then rediscover that same relationship. The result would prove nothing.

The correction. The voice-command dataset (Layer B) will be built from real recordings rather than generated text. The plan is roughly 25 to 30 speakers, each recording 10 to 12 payment commands on a mobile microphone in ordinary room conditions, deliberately including older and non-native English speakers, since they are the users the project is aimed at. Those recordings are passed through Whisper and Google Speech-to-Text, and the confidence values the engines actually return become the features. Each recording is then paired with a simulated transaction context using a documented pairing procedure. The transaction side may be simulated — no student project can obtain real bank records — but the voice side must be real.

Supporting evidence from the literature. This approach follows accepted practice. VISHGUARD, published in Nature Scientific Data, is a corpus of 3,000 simulated fraud phone calls. TeleAntiFraud-28k (ACM Multimedia 2025) combines simulated and real audio across 28,511 audio-text pairs. What makes such datasets publishable is not that the data is real, but that the construction procedure is documented, the labels are checked by a second annotator with an agreement score reported, and the limitation is stated in the paper.

Additional outcome. No public dataset of real UPI voice-payment commands with risk labels currently exists. Building one has therefore been moved from "future work" into a stated contribution of this project.

13. Backend Progress

The Milestone-1 backend remains stable. The FastAPI server runs with the /health endpoint and the logging system is in place. The configuration files created in Milestone 1 (thresholds.py, weights.py) have been retained; thresholds.py now holds the cut-offs as clearly marked placeholder values pending the calibration described in Section 9, and weights.py serves the rule-based baseline only.

[Insert Fig. 1 — screenshot of your repository / updated folder structure, then delete this box]

[Insert Fig. 2 — screenshot of code written this week, e.g. thresholds.py with the placeholder comment, then delete this box]

Note: *this was primarily a research and design milestone, since the mentor's action items from Milestone 1 were research questions. Implementation of the ML pipeline begins in Milestone 3.*

14. This Week's Learning — Completed

- How machine-learning models are used for fraud and risk scoring, and why gradient boosting is the usual choice on tabular data.
- Feature engineering: turning raw transaction and voice data into usable numeric features.

- Handling severe class imbalance using class weighting and resampling (imbalanced-learn).
- SHAP, and how feature attributions are computed and read.
- Probability calibration, and why a model score is not automatically a probability.
- Why PR-AUC and Recall matter more than plain Accuracy when only 0.13% of records are fraud.
- Reading and verifying research papers, including checking that a cited source actually exists.

15. Updated Technology Stack

Flutter, FastAPI, Python, Gemini / OpenAI (LLM), Speech-to-Text, **scikit-learn, XGBoost, SHAP, imbalanced-learn, pandas**, Retrieval-Augmented Generation (RAG), Vector Database, UPI Deep Linking. Additions from this milestone are shown in bold.

16. Meeting with Mentor and Things to Work On

All three action items from Milestone 1 have been completed and are documented above. Two points require the mentor's input before Milestone 3 begins:

1. **Primary model choice.** Random Forest trains faster and is far less sensitive to hyperparameter tuning, which is safer given the November deadline. XGBoost is likely to score slightly higher but needs more tuning time. Confirmation is requested on which should be treated as the primary model, with the other kept as a comparison.
2. **Voice data collection.** Approval is requested for recording payment commands from roughly 25 to 30 volunteer speakers, along with a consent form, as described in Section 12.

Planned work for the next milestone:

- Write the recording protocol: the command templates, speaker targets, and the Safe / Ambiguous / Risky labelling rules with a second-annotator check.
- Prepare the consent form and begin voice data collection.
- Download PaySim and IEEE-CIS and complete initial data exploration and cleaning.
- Build the feature extraction module for the transaction and behaviour categories.
- Train the first baseline models (rule-based and Logistic Regression) to establish a comparison point.

17. Milestone Status

> **Milestone 2 has been successfully completed.** All three mentor action items were addressed. The Risk Module has been redesigned from a fixed rule-based score to a trained, explainable machine-learning model; the risk signal set has been expanded from 5 signals to about 25 features across four categories; the threshold values are now derived from data instead of chosen by hand; and the research contribution has been narrowed to a single testable claim with an experiment designed to test it. The revised research proposal (VeriPay CPDE v2) is submitted alongside this report.